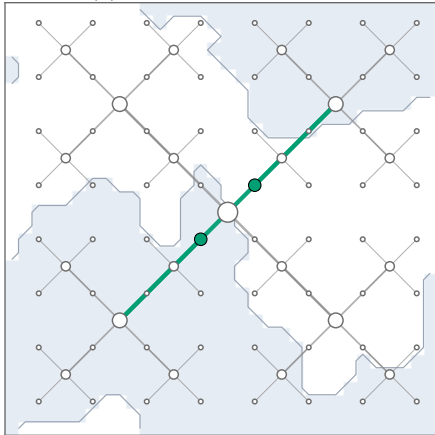
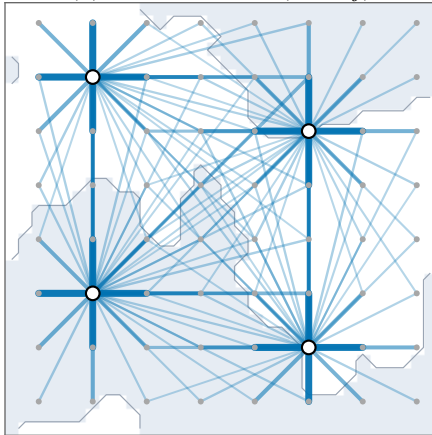


(a) tree: U-Net hierarchy



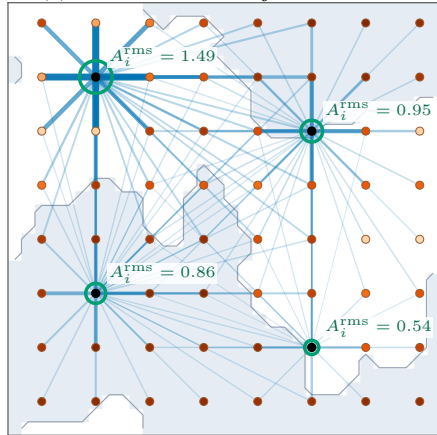
neighbouring cells across a block
boundary meet only at the root

(b) global, fixed: $\kappa_{\theta}(\mathbf{r}_i - \mathbf{r}_j)$



every cell hears every other through
the same pattern (Fourier layer)

(c) rhizome: $\mathbf{A}_i \kappa_{\theta} \mathbf{B}_j$, update $t=4$



edge $j \rightarrow i$ scaled by the receiver gate
 $\mathbf{A}_i$ (ring) and the sender gate $\mathbf{B}_j$ (shade)